MIL-F-83860/4(USAF)
17 FEB 86

MILITARY SPECIFICATION SHEET

FILTER ELEMENT, NONCLEANABLE, FLUID PRESSURE, HYDRAULIC LINE, 5 MICRON ABSOLUTE, TYPE II SYSTEMS

This specification is approved for use within the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

The requirements for acquiring the filter element described herein shall consist of this specification and the latest issue of MIL-F-83860.

REQUIREMENTS:

BUBBLE POINT. The initial and burst out bubble point values shall meet the values established by the activity responsible for qualification and shall be determined as specified in 4.7.1.1 of general specification.

FILTRATION. The filter element shall remove 100 percent of all contaminants over 5 microns in size and 95 percent by weight of add contaminants.

COLD START. The filter elements shall withstand 3000 psid differential pressure across the element for the flow cycles and fluid temperature as specified in 4.7.3 of general specification.

ELEMENT PRESSURE DROP. The pressure drop through the clean element at rated flow shall not exceed the values specified. Test fluid shall be maintained at 100 ± 10 F.

Part Number	Rated Flow	Pressure Drop
M83860/4-1	24 gpm	18 psid
M83860/4-2	3 gpm	.8 psid
M83860/4-3	12 gpm	8 psid

ELEMENT DIRT CAPACITY. The filter element shall meet the dirt capacities specified in Table I, Contaminants.

ELEMENT COLLAPSE. The filter element shall withstand differential pressures specified:

M83860/4-1	4500 psid
M83860/4-2	150 psid
M83860/4-3	150 psid

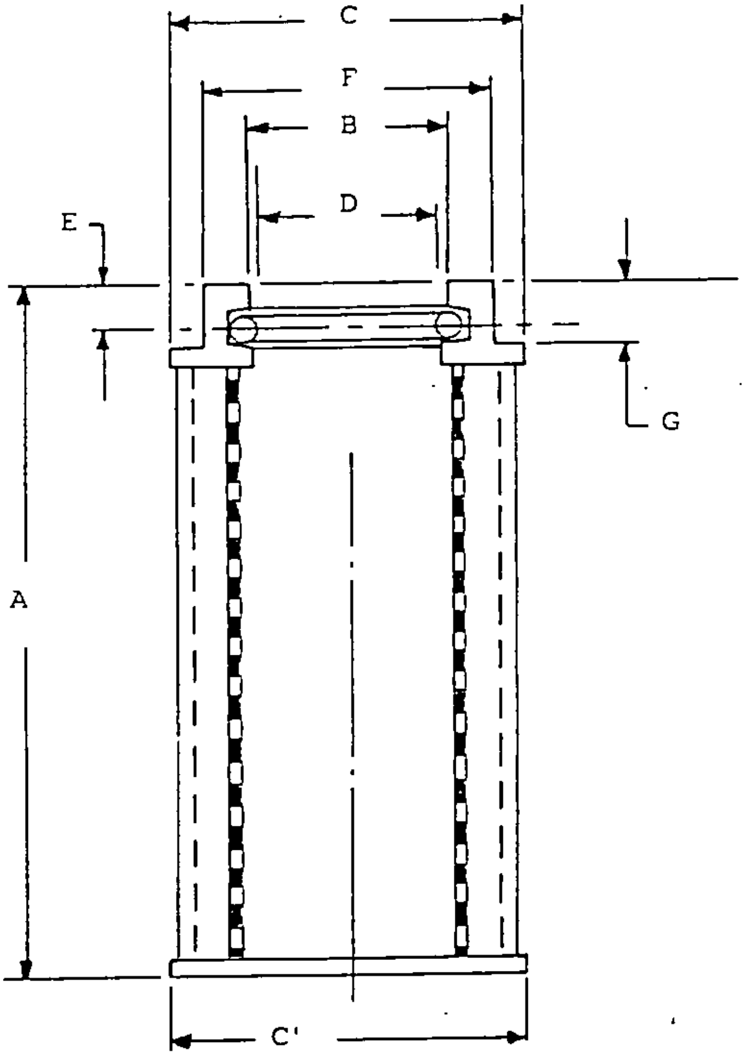
DESCRIPTION. The filter element shall be as shown in Figure 1 and as specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1650

DISTRIBUTION STATEMENT. Approved for public release; distribution is unlimited.



DASH No.	RATED Flow gpm	A +.062 - 0	B ±.015	C&C ±.015	D NOM	E MAX	F ±.015	G ±.015	"O" RING
-1	24	4.437	1.015	1.750	1.000	.218	-	-	MS9388-214
-2	3	4.625	.640	1.125	.625	.125	-	-	MS28775-114
-3	12	4.437	1.015	1.750	1.000	.218	-	-	MS9388-214

FIGURE 1

FIGURE 1. Filter elements, nonclearable, fluid pressure, hydraulic line, 5 micorn absolute, type II system.

FLOW FATIGUE. The filter elements shall withstand the flow fatigue cycles specified at 300 cpm:

Cycles	Pressure Drop (PSID)
1,120,000	20
112,000	50
12,500	100
400	1500
100	3000

MEDIA MIGRATION. Element migration when tested in accordance with 4.7.8.3.2 of general specification shall not exceed 0.0040 grams and there shall be no migration identifiable as coming from the filter media.

MAXIMUM PARTICLE PASSED. The maximum particle passed shall be no greater than 5 microns.

TABLE I. CONTAMINANTS

		Degree of Filtration Test		Maximum Particle Test	Pressure Buildup Test	
DASH No.	FLOW Rate (gpm)	ADD RATE F-9 Beads (gpms)	ADD RATE AC Dust (gpms)	ADD RATE Carbony 1 (gpms)	ADD RATE AC Dust (gpms)	MINIMUM AC Dust (gpms)
-1	24	0.15	0.15	0.30	0.25	4.5 (90 psid)
-2	3	0.15	0.15	0.30	0.25	4.5 (40 psid)
-3	12	0.15	0.15	0.30	0.50	7.0 (40 psid)

PART NUMBER. The part number consists of an "M: plus the basic number of this specification sheet and a dash number.

EXAMPLE: M83860/4-1

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 71

Project Number:
1650 - F138